

ORIGINAL RESEARCH

HEART FAILURE

# Diet and Clinical Outcomes in a Heart Failure Population



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## ABSTRACT

**BACKGROUND** There are limited data to inform dietary recommendations in persons with heart failure (HF).

**OBJECTIVES** The aim of this study was to examine associations between consumption levels of 11 common foods, and a healthy diet pattern, with clinical outcomes in a HF population.

**METHODS** This analysis was a substudy of 3,798 participants from 25 countries enrolled in the multinational G-CHF (Global Congestive Heart Failure) registry with dietary data collected through food frequency questionnaires. Associations were examined between consumption levels of 6 plant-based foods (fruits, vegetables, legumes, nuts, whole grains, and refined grains) and 5 animal-based foods (fish, poultry, unprocessed red meat, eggs, and dairy) with the primary composite outcome of death or HF hospitalization and its components. Also examined was the association between an overall healthy diet pattern (measured by using a mAHEI [modified Alternative Healthy Eating Index]) and these outcomes.

**RESULTS** A total of 1,236 participants had a primary outcome event, 890 participants died, and 593 were hospitalized for HF. Higher legume intake (HR: 0.85 [95% CI: 0.73-0.99] for 0.1 to <0.5 serving per day and HR: 0.80 [95% CI: 0.65-0.98] for ≥0.5 serving per day vs <0.1 serving per day) was associated with a lower risk of the primary outcome. Moderate vegetable intake (1 to <3 servings per day) was associated with a lower risk of HF hospitalization (HR: 0.77 [95% CI: 0.61-0.97]) compared with <1 serving per day. Higher refined grain intake was associated with a higher risk of hospitalization for HF (HR: 1.56 [95% CI: 1.19-2.05] for 1-3 servings per day and HR: 1.76 [95% CI: 1.30-2.39] for >3 servings per day vs <1 serving per day). Associations with other foods, as well as with the mAHEI, were neutral.

**CONCLUSIONS** In persons with HF, higher legume and vegetable intake were each associated with a lower risk of adverse clinical outcomes, whereas higher refined grain intake was associated with a higher risk of adverse clinical outcomes. (JACC Heart Fail. 2026;14:102728) © 2026 by the American College of Cardiology Foundation.

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**ABBREVIATIONS  
AND ACRONYMS****BMI** = body mass index**FFQ** = food frequency  
questionnaire**HF** = heart failure

Worldwide, heart failure (HF) affects approximately 64 million people.<sup>1</sup> Despite medical advances, HF is still associated with high morbidity and mortality.<sup>2</sup> Although there are robust data from clinical trials to advise guidance on sodium consumption in persons with HF, data regarding the impact of other dietary components on HF outcomes are limited, and it is unclear whether persons with HF should follow guidelines for a healthy diet as outlined for the general population, or whether specific considerations are needed.<sup>3-5</sup> Diet can have downstream effects on pathways that affect HF progression, including neurohormonal activation, cardiac energetics, and cachexia.<sup>6-9</sup> Data informing the optimal food choices for patients with HF remain an acknowledged evidence gap.

The G-CHF (Global Congestive Heart Failure) study is a multinational registry of 23,341 patients with HF recruited from 40 countries. In a substudy of 3,798 participants from 25 countries, food frequency questionnaires (FFQs) were administered at baseline, allowing for a comprehensive assessment of habitual dietary intake. We examined relationships between consumption levels of 11 common food groups and death or hospitalization for HF. We also examined whether a healthy diet pattern for the general population was associated with clinical outcomes.

**METHODS**

**STUDY DESIGN AND PARTICIPANTS.** The design of the G-CHF registry has been reported previously.<sup>2,10</sup> It is an international, multicenter registry of patients with HF. The study enrolled 23,341 adult patients with a diagnosis of HF from 40 countries. Recruitment of participants from both outpatient and inpatient hospital settings occurred between 2016 and 2020. In a dietary substudy, habitual dietary intake was measured in 3,798 individuals from 25 countries: Canada, the United States, Argentina, Brazil, Chile, Colombia, Ecuador, Germany, Italy, Poland, Portugal,

Spain, Sweden, Russia, Botswana, Cameroon, Kenya, Mozambique, Nigeria, Uganda, Saudi Arabia, Turkey, China, India, and Nepal. For this analysis, countries were grouped into 4 geographical regions (South America, Asia, Africa, and the region consisting of North America, Europe, and Russia) to reflect the geographical diversity of the cohort but to also allow for a sufficient number of participants in each region for analyses.

Central ethics approval for the study was obtained through the Hamilton Integrated Research Ethics Board. Local ethics approvals were also obtained by ethics committees at all participating centers for sites within each country. All participants provided informed written consent before taking part in the study.

**MEASUREMENT OF DIET.** At baseline, FFQs were used to collect habitual food and beverage intake. Country- and region-specific FFQs previously validated in the multinational, community-based PURE (Prospective Urban Rural Epidemiology) study were adapted by shortening the food lists to include the food and beverage items most commonly consumed by participants within each country.<sup>11</sup> The food and beverage lists for each FFQ included between 47 and 89 items. For each food or beverage, we asked the frequency of consumption of a specified portion size, varying from never to >3 times per day, were used. Participants were asked “In the last 12 months, how often did you eat foods from each of the following categories?” and the list of food items was given with a specified portion size. Daily intake of each food was calculated by multiplying the portion size and daily frequency of consumption, and this was then converted to servings per day using standard serving portions. Using this approach, consumption levels of 6 common plant-based foods (fruits, vegetables, legumes, nuts, whole grains, and refined grains) and 5 animal-based foods (fish, poultry, unprocessed red meat, eggs, and dairy) were estimated for each participant.

Our approach for measuring overall diet quality was based on an adaption of the AHEI (Alternative

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The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

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Healthy Eating Index) 2010 focused on its food and nutrient components.<sup>12,13</sup> A healthy diet as measured by the AHEI-2010 is associated with a lower risk of cardiovascular disease in studies of the general population.<sup>12</sup> Our mAHEI (modified Alternative Healthy Eating Index) included 6 food or nutrient groups in which higher intake is considered to be better for health (fruits, vegetables, whole grains, nuts and legumes, polyunsaturated fatty acids, and long-chain omega-3 fats) and 2 food groups in which higher intake is considered to be worse for health (unprocessed red meats and soft drinks). Higher scores were given to higher consumption of healthy foods and lower consumption of unhealthy foods. Points were also proportionally assigned for intermediate intake of food items. Each food group was scored on a 10-point scale, with the total score ranging from a value of 0 to 80 points, and higher scores indicating greater adherence to a healthy diet (Supplemental Table 1). Six variables from our FFQ (ie, fruits, vegetables, whole grains, nuts and legumes, unprocessed red meats, soft drinks) were directly incorporated into the score. Polyunsaturated fatty acid and long-chain omega-3 fat intake was estimated by first calculating fish and nut consumption (which were the major dietary sources in our data set) in participants by using the FFQs, and then relating consumption levels of these items to the United States Department of Agriculture national nutrient database. Although sodium and trans fatty acids intake are included in the AHEI-2010, these could not be estimated by using the study FFQs.

**SELECTION AND COLLECTION OF BASELINE COVARIATES.** Additional covariates were selected for analyses to account for socioeconomic status, enrollment as an inpatient or outpatient, geographical region, and clinical risk factors associated with adverse HF outcomes. Clinical characteristics were chosen to be consistent with those in the MAGGIC (Meta-Analysis Global Group in Chronic) HF risk score, which represents a parsimonious set of variables that have been associated with mortality in patients with HF.<sup>14,15</sup> We defined risk factors at baseline in the same manner as the main registry, including self-reported smoking, history of hypertension, diabetes mellitus, and chronic obstructive pulmonary disease.<sup>2</sup> Education level was collected as the primary socioeconomic variable of interest. Baseline measurements were taken for height and weight (to calculate body mass index [BMI]) and for systolic blood pressure. Stages of chronic kidney disease were defined based on estimated glomerular filtration rate using local laboratory measurements.

Echocardiographic data on ejection fraction were collected if an echocardiogram was performed within 1 year of enrollment; using this test, participants were classified as having heart failure with preserved, mildly reduced, or reduced ejection fraction. Investigator-reported HF etiology, medication use, having an implantable cardioverter-defibrillator, and self-reported consumption of a low-salt diet were also collected at baseline and used as covariates for the current analysis.

**FOLLOW-UP AND CLINICAL OUTCOMES.** Clinical outcomes were collected by using a standardized questionnaire every 6 months during the first 2 years of the study and yearly afterward as part of the study's long-term extension. Deaths and hospitalizations were collected at each scheduled study visit. The primary outcome was the composite of time to death from any cause or first HF hospitalization. As secondary outcomes, we examined each component of the primary outcome.

**STATISTICAL ANALYSIS.** Standard descriptive statistical methods were used: mean  $\pm$  SD or median (Q1-Q3) for continuous data, and the absolute and percentages for categorical data. For all food items, the extreme values of consumption were truncated to the estimated 99th percentile of consumption. First, we evaluated associations between the levels of consumption of each food group and the primary composite outcome, as well as each component of this outcome. Next, we assessed the association between the mAHEI and these outcomes. The mAHEI was examined as a categorical variable stratified according to quartiles. Cox regression models were used to calculate HRs and 95% CIs. The proportional hazards assumption was assessed by visually inspecting log-log plots of the Kaplan-Meier survival curves. Models were adjusted for age and sex and for the following covariates: recruitment type (inpatient/outpatient), level of education (primary school or less, secondary school, postsecondary school), smoking (no, yes, former), BMI, history of diabetes, chronic obstructive pulmonary disease, medication use, systolic blood pressure, time since HF diagnosis, stage 4 or 5 kidney disease based on estimated glomerular filtration rate, HF type, NYHA functional class, HF etiology, geographic region, self-reported adherence to a low-salt diet (as a yes or no response), alcohol intake, and additional dietary variables. For each category of food consumption, a value of  $P < 0.05$  was considered statistically significant.

Analyses were conducted with R version 4.4.1 (R Foundation for Statistical Computing).

**TABLE 1** Baseline Clinical Characteristics and Food Intake Patterns of Participants Overall and According to Geographical Region

|                                                | Overall<br>(N = 3,798) | Africa<br>(n = 638) | Asia<br>(n = 912)   | South America<br>(n = 676) | North America,<br>Europe, and Russia<br>(n = 1,572) |
|------------------------------------------------|------------------------|---------------------|---------------------|----------------------------|-----------------------------------------------------|
| <b>Demographic characteristics</b>             |                        |                     |                     |                            |                                                     |
| Age, y                                         | 61.9 ± 14.1            | 49.9 ± 15.8         | 57.1 ± 12.8         | 65.9 ± 11.2                | 67.9 ± 10.9                                         |
| Male                                           | 2,478 (65.2)           | 280 (43.9)          | 610 (66.9)          | 442 (66.1)                 | 1,137 (72.6)                                        |
| Inpatient                                      | 762 (20.1)             | 115 (18.0)          | 238 (26.1)          | 30 (4.5)                   | 378 (24.1)                                          |
| <b>Education</b>                               |                        |                     |                     |                            |                                                     |
| Primary or less                                | 1,477 (38.9)           | 271 (42.5)          | 481 (52.7)          | 378 (56.5)                 | 341 (21.8)                                          |
| Secondary                                      | 1,290 (34.0)           | 218 (34.2)          | 260 (28.5)          | 201 (30.0)                 | 606 (38.8)                                          |
| Postsecondary                                  | 1,027 (27.1)           | 149 (23.4)          | 171 (18.8)          | 90 (13.5)                  | 616 (39.4)                                          |
| <b>Medical conditions and health behaviors</b> |                        |                     |                     |                            |                                                     |
| Diabetes                                       | 1,112 (29.3)           | 52 (8.2)            | 320 (35.1)          | 226 (33.8)                 | 509 (32.5)                                          |
| Systolic blood pressure, mm Hg                 | 123 ± 20               | 122 ± 23            | 122 ± 19            | 124 ± 21                   | 124 ± 18                                            |
| Body mass index, kg/m <sup>2</sup>             | 28 ± 6                 | 26 ± 6              | 25 ± 5              | 29 ± 6                     | 30 ± 6                                              |
| Current smoker                                 | 359 (9.5)              | 6 (0.9)             | 106 (11.6)          | 30 (4.5)                   | 215 (13.7)                                          |
| <b>HF features and treatment</b>               |                        |                     |                     |                            |                                                     |
| HF diagnosis >12 mo prior                      | 2,826 (74.4)           | 376 (58.9)          | 639 (70.1)          | 568 (84.9)                 | 1,235 (78.8)                                        |
| <b>Left ventricular ejection fraction</b>      |                        |                     |                     |                            |                                                     |
| ≥50%                                           | 740 (23.7)             | 129 (22.4)          | 110 (14.4)          | 128 (23.5)                 | 373 (30.3)                                          |
| 40%-49%                                        | 527 (16.9)             | 70 (12.2)           | 115 (15.1)          | 107 (19.7)                 | 233 (18.9)                                          |
| ≤40%                                           | 1,850 (59.4)           | 376 (65.4)          | 539 (70.5)          | 309 (56.8)                 | 624 (50.7)                                          |
| <b>NYHA functional class</b>                   |                        |                     |                     |                            |                                                     |
| I                                              | 563 (14.9)             | 122 (19.1)          | 84 (9.2)            | 88 (13.2)                  | 269 (17.3)                                          |
| II                                             | 2,196 (58.0)           | 290 (45.5)          | 518 (56.8)          | 417 (62.3)                 | 960 (61.6)                                          |
| III                                            | 889 (23.5)             | 159 (25.0)          | 271 (29.7)          | 149 (22.3)                 | 309 (19.8)                                          |
| IV                                             | 142 (3.7)              | 67 (10.5)           | 39 (4.3)            | 15 (2.2)                   | 21 (1.3)                                            |
| Beta-blocker                                   | 3,167 (83.4)           | 442 (69.3)          | 747 (81.9)          | 586 (87.7)                 | 1,381 (88.1)                                        |
| Renin-angiotensin-aldosterone system inhibitor | 3,064 (80.7)           | 528 (82.8)          | 632 (69.3)          | 561 (84.0)                 | 1,332 (85.0)                                        |
| <b>Dietary intake</b>                          |                        |                     |                     |                            |                                                     |
| Fruits, servings/d                             | 0.85 (0.36-1.38)       | 0.56 (0.19-1.19)    | 0.49 (0.17-1.13)    | 0.84 (0.36-1.17)           | 0.97 (0.45-2.28)                                    |
| Vegetables, servings/d                         | 1.60 (0.90-2.86)       | 1.48 (0.78-2.81)    | 2.71 (1.45-4.71)    | 1.40 (0.90-2.30)           | 1.38 (0.76-2.39)                                    |
| Legumes, servings/d                            | 0.19 (0.07-0.48)       | 0.11 (0.05-0.32)    | 0.49 (0.25-0.95)    | 0.35 (0.12-0.86)           | 0.11 (0.05-0.24)                                    |
| Refined grains, servings/d                     | 1.63 (0.75-3.05)       | 1.05 (0.56-2.62)    | 0.71 (0.37-1.86)    | 2.22 (1.40-3.48)           | 2.05 (1.13-3.52)                                    |
| Whole grains, servings/d                       | 0.62 (0.07-1.93)       | 2.03 (0.69-4.09)    | 0.86 (0.17-2.64)    | 0.04 (0.00-0.72)           | 0.48 (0.12-1.26)                                    |
| Nuts, servings/d                               | 0.08 (0.00-0.31)       | 0.31 (0.08-1.11)    | 0.09 (0.00-0.25)    | 0.00 (0.00-0.00)           | 0.07 (0.00-0.28)                                    |
| Unprocessed red meat, servings/d               | 0.71 (0.28-1.54)       | 1.35 (0.44-3.30)    | 0.18 (0.00-0.58)    | 1.20 (0.54-1.66)           | 0.71 (0.37-1.46)                                    |
| Poultry, servings/d                            | 0.51 (0.26-1.03)       | 0.30 (0.14-0.90)    | 0.37 (0.10-0.81)    | 1.17 (0.40-2.07)           | 0.59 (0.30-1.03)                                    |
| Fish, servings/d                               | 0.14 (0.07-0.43)       | 0.14 (0.07-0.43)    | 0.07 (0.00-0.14)    | 0.14 (0.07-0.37)           | 0.21 (0.14-0.43)                                    |
| Eggs, servings/d                               | 0.39 (0.13-0.51)       | 0.14 (0.07-0.43)    | 0.28 (0.07-0.91)    | 0.43 (0.14-0.45)           | 0.39 (0.13-0.67)                                    |
| Dairy, servings/d                              | 0.88 (0.26-1.60)       | 0.32 (0.10-0.86)    | 1.07 (0.31-1.64)    | 0.87 (0.26-1.49)           | 1.06 (0.46-1.77)                                    |
| Modified AHEI score                            | 38.00 (29.00-46.00)    | 38.00 (30.00-49.00) | 47.00 (39.00-52.00) | 32.50 (25.00-40.00)        | 35.00 (28.00-42.00)                                 |

Values are mean ± SD, n (%), or median (Q1-Q3).  
AHEI = Alternative Healthy Eating Index.

## RESULTS

During the study period of enrollment from December 2016 to March 2020, the baseline dietary intake of 3,798 participants from 25 countries was recorded. Of these, 3,788 (99.7%) had at least one follow-up visit available to document clinical events. Median follow-up of the study cohort included in this

analysis was 3 years. During follow-up, 1,236 had the composite outcome, 890 participants died, and 593 participants reported a hospitalization for HF.

**Table 1** shows the baseline characteristics and food consumption patterns of participants overall and stratified according to geographical region. A total of 638 (16.8%) participants were from Africa, 912 (23.4%) were from Asia, 676 (24.0%) were from South

**TABLE 2 Associations Between Plant-Based Food Groups and Clinical Outcomes**

| Categories of Intake and Crude Numbers of Events |                         |                                  |                       | Adjusted HR (95% CI) <sup>a</sup> |                               |                               |
|--------------------------------------------------|-------------------------|----------------------------------|-----------------------|-----------------------------------|-------------------------------|-------------------------------|
|                                                  | <1 serv/d (n = 2,135)   | 1 to <2 serv/d (n = 850)         | ≥2 serv/d (n = 813)   | <1 serv/d                         | 1 to <2 serv/d                | ≥2 serv/d                     |
| <b>Fruits</b>                                    |                         |                                  |                       |                                   |                               |                               |
| Death or HF hospitalization                      | 630                     | 277                              | 240                   | 1                                 | 1.04 (0.88-1.23)              | 0.93 (0.78-1.13)              |
| Death                                            | 493                     | 221                              | 175                   | 1                                 | 1.04 (0.86-1.24)              | 0.94 (0.75-1.17)              |
| HF hospitalization                               | 258                     | 109                              | 106                   | 1                                 | 1.13 (0.88-1.45)              | 0.86 (0.65-1.14)              |
| <b>Vegetables</b>                                | <1 serv/d (n = 1,113)   | 1 to <3 serv/d (n = 1,825)       | ≥3 serv/d (n = 860)   | <1 serv/d                         | 1 to <3 serv/d                | ≥3 serv/d                     |
| Death or HF hospitalization                      | 362                     | 524                              | 261                   | 1                                 | 0.89 (0.77-1.04)              | 0.85 (0.69-1.05)              |
| Death                                            | 274                     | 412                              | 203                   | 1                                 | 1.00 (0.84-1.19)              | 0.90 (0.71-1.15)              |
| HF hospitalization                               | 169                     | 208                              | 96                    | 1                                 | 0.77 (0.61-0.97) <sup>b</sup> | 0.78 (0.56-1.08)              |
| <b>Legumes</b>                                   | <0.1 serv/d (n = 1,191) | <0.1-0.5 serv/d (n = 1,685)      | ≥0.5 serv/d (n = 922) | <0.1 serv/d                       | <0.1-0.5 serv/d               | ≥0.5 serv/d                   |
| Death or HF hospitalization                      | 400                     | 496                              | 251                   | 1                                 | 0.85 (0.73-0.99) <sup>b</sup> | 0.80 (0.65-0.98) <sup>b</sup> |
| Death                                            | 302                     | 379                              | 208                   | 1                                 | 0.87 (0.73-1.04)              | 0.91 (0.72-1.16)              |
| HF hospitalization                               | 187                     | 207                              | 79                    | 1                                 | 0.84 (0.67-1.05)              | 0.62 (0.44-0.87) <sup>c</sup> |
| <b>Refined grains</b>                            | <1 serv/d (n = 1,260)   | 1-3 serv/d (n = 1,565)           | ≥3 serv/d (n = 973)   | <1 serv/d                         | 1 to 3 serv/d                 | ≥3 serv/d                     |
| Death or HF hospitalization                      | 409                     | 461                              | 277                   | 1                                 | 1.07 (0.91-1.26)              | 1.15 (0.95-1.39)              |
| Death                                            | 352                     | 338                              | 199                   | 1                                 | 0.92 (0.76-1.12)              | 0.99 (0.79-1.23)              |
| HF hospitalization                               | 118                     | 219                              | 136                   | 1                                 | 1.56 (1.19-2.05) <sup>d</sup> | 1.76 (1.30-2.39) <sup>d</sup> |
| <b>Whole grains</b>                              | <0.1 serv/d (n = 1,035) | 0.1-1 serv/d (n = 1,257)         | >1 serv/d (n = 1,506) | <0.1 serv/d                       | 0.1-1 serv/d                  | >1 serv/d                     |
| Death or HF hospitalization                      | 301                     | 369                              | 477                   | 1                                 | 1.02 (0.85-1.23)              | 1.06 (0.88-1.28)              |
| Death                                            | 246                     | 280                              | 363                   | 1                                 | 0.98 (0.80-1.21)              | 0.97 (0.79-1.20)              |
| HF hospitalization                               | 117                     | 163                              | 193                   | 1                                 | 1.00 (0.76-1.32)              | 1.05 (0.79-1.39)              |
| <b>Nuts</b>                                      | <1 serv/wk (n = 1,892)  | 1 serv/wk to 1 ser/d (n = 1,070) | >1 serv/d (n = 462)   | <1 serv/wk                        | 1 serv/wk to 1 serv/d         | >1 serv/d                     |
| Death or HF hospitalization                      | 568                     | 334                              | 147                   | 1                                 | 0.96 (0.81-1.13)              | 0.87 (0.69-1.10)              |
| Death                                            | 439                     | 247                              | 120                   | 1                                 | 0.87 (0.72-1.06)              | 0.98 (0.75-1.28)              |
| HF hospitalization                               | 240                     | 153                              | 50                    | 1                                 | 1.18 (0.92-1.50)              | 0.76 (0.52-1.11)              |

<sup>a</sup>Multivariable analyses were adjusted for age, sex, inpatient recruitment, body mass index, education, current smoker, alcohol intake, NYHA functional class, history of diabetes, history of chronic obstructive pulmonary disease, beta-blocker use, renin-angiotensin system inhibitor use, mineralocorticoid receptor antagonist use, implantable cardioverter-defibrillator use, stage 4 or 5 kidney disease, left ventricular ejection fraction category, heart failure (HF) etiology, systolic blood pressure, self-reported low-salt diet, and geographical region. Analyses were also adjusted for the following dietary variables (measured in grams per day): For associations with fruit intake, models are adjusted for vegetables, legumes, starches, and unprocessed red meats. For vegetables, models are adjusted for fruits, starches, and unprocessed red meats. For legumes, models are adjusted for fruits, starches, and unprocessed red meats. For refined grains, models are adjusted for fruits, vegetables, legumes, and unprocessed red meats. For nuts, models are adjusted for fruits, vegetables, legumes, starches, and unprocessed red meats. <sup>b</sup>Statistically significant associations at a threshold value of  $P < 0.05$ . <sup>c</sup>Statistically significant associations at a threshold value of  $P < 0.01$ . <sup>d</sup>Statistically significant associations at a threshold value of  $P < 0.001$ .

serv/d = servings per day; serv/wk = servings per week.

America, and 1,572 (41.4%) participants were from the combined region of North America, Europe, and Russia.

The mean age of the population was  $61.9 \pm 14.1$  years; 2,478 (65.2%) participants were male, and 762 (20.1%) participants were enrolled from the hospital inpatient setting. Left ventricular ejection fraction was  $\leq 40\%$  in 1,850 (59.4%) participants, between 41% and 49% in 527 (16.9%) participants, and  $\geq 50\%$  in 740 (23.7%) participants. A total of 1,031 (27.2%) participants had NYHA functional class III or IV symptoms. The median mAHEI score was 38 (Q1-Q3: 29-46). The combined region of North America, Europe, and Russia had the highest consumption levels of fruits and fish. South America had the highest consumption levels of refined grains, poultry, and eggs. Asia had the highest consumption levels of vegetables, legumes, and dairy. Africa had the highest consumption levels of whole grains, nuts, and unprocessed red meat. The highest diet quality,

as measured by the mAHEI, was observed in Asia and the lowest in South America. Correlations between consumption levels of individual food groups were small to moderate in size ([Supplemental Table 2](#)).

Associations between each food group and outcomes are presented in [Tables 2 and 3](#). Of the plant-based foods, higher consumption of legumes was associated with a significantly lower risk of the primary outcome (HR: 0.85 [95% CI: 0.73-0.99] for 0.1 to 0.5 serving per day and HR: 0.80 [95% CI: 0.65-0.98] for  $\geq 0.5$  serving per day vs  $<0.1$  serving per day). This effect was predominantly driven by the association between higher legume intake and a lower risk of first hospitalization for HF (HR: 0.84 [95% CI: 0.67-1.05] for 0.1 to  $<0.5$  serving per day and HR: 0.62 [95% CI: 0.44-0.87] for  $\geq 0.5$  serving per day vs  $<0.1$  serving per day). Higher levels of vegetable intake were not associated with the primary outcome, but moderate vegetable intake (1 to  $<3$  servings per day) was associated with a lower risk of HF hospitalization

**TABLE 3 Associations Between Animal-Based Food Groups and Clinical Outcomes**

|                             | Categories of Intake and Crude Numbers of Events |                                       |                            | Adjusted HR (95% CI) <sup>a,b</sup> |                        |                  |
|-----------------------------|--------------------------------------------------|---------------------------------------|----------------------------|-------------------------------------|------------------------|------------------|
|                             |                                                  |                                       |                            |                                     |                        |                  |
| Fish                        | <1 serv/wk<br>(n = 1,934)                        | 1 to 3.5 serv/wk<br>(n = 1,117)       | ≥3.5 serv/wk<br>(n = 446)  | <1 serv/wk                          | 1 to 3.5 serv/wk       | ≥3.5 serv/wk     |
| Death or HF hospitalization | 579                                              | 362                                   | 124                        | 1                                   | 1.10 (0.94-1.29)       | 1.01 (0.79-1.28) |
| Death                       | 471                                              | 272                                   | 92                         | 1                                   | 1.07 (0.89-1.29)       | 0.95 (1.02-1.26) |
| HF hospitalization          | 223                                              | 152                                   | 49                         | 1                                   | 1.08 (0.84-1.38)       | 0.97 (0.67-1.41) |
| Unprocessed red meat        | ≤2 serv/wk<br>(n = 1,949)                        | 2 serv/wk to <1 serv/d<br>(n = 1,292) | ≥1 serv/d<br>(n = 557)     | ≤2 serv/wk                          | 2 serv/wk to <1 serv/d | ≥1 serv/d        |
| Death or HF hospitalization | 606                                              | 390                                   | 151                        | 1                                   | 0.97 (0.84-1.13)       | 1.05 (0.85-1.30) |
| Death                       | 476                                              | 302                                   | 111                        | 1                                   | 0.90 (0.75-1.08)       | 0.93 (0.73-1.19) |
| HF hospitalization          | 246                                              | 166                                   | 61                         | 1                                   | 0.99 (0.78-1.25)       | 1.06 (0.76-1.48) |
| Poultry                     | <3 serv/wk<br>(n = 1,191)                        | 3 serv/wk to 1 serv/d<br>(n = 1,410)  | ≥1 serv/d<br>(n = 1,197)   | <3 serv/wk                          | <1 serv/d              | ≥1 serv/d        |
| Death or HF hospitalization | 378                                              | 453                                   | 316                        | 1                                   | 1.06 (0.90-1.24)       | 1.03 (0.86-1.24) |
| Death                       | 238                                              | 281                                   | 195                        | 1                                   | 1.02 (0.85-1.23)       | 1.08 (0.87-1.33) |
| HF hospitalization          | 147                                              | 197                                   | 129                        | 1                                   | 1.18 (0.92-1.51)       | 0.99 (0.75-1.32) |
| Eggs                        | <0.2 serv/d<br>(n = 1,713)                       | 0.2-0.5 serv/d<br>(n = 1,032)         | ≥0.5 serv/d<br>(n = 1,053) | <0.2 serv/d                         | 0.2-0.5 serv/d         | ≥0.5 serv/d      |
| Death or HF hospitalization | 561                                              | 267                                   | 319                        | 1                                   | 0.90 (0.76-1.06)       | 0.99 (0.83-1.17) |
| Death                       | 438                                              | 209                                   | 242                        | 1                                   | 0.92 (0.76-1.12)       | 1.05 (0.86-1.29) |
| HF hospitalization          | 233                                              | 113                                   | 127                        | 1                                   | 0.92 (0.72-1.19)       | 0.86 (0.66-1.13) |
| Dairy                       | <0.5 serv/d<br>(n = 1,303)                       | 0.5-1.5 serv/d<br>(n = 1,452)         | >1.5 serv/d<br>(n = 1,043) | <0.5 serv/d                         | 0.5-1.5 serv/d         | >1.5 serv/d      |
| Death or HF hospitalization | 395                                              | 434                                   | 318                        | 1                                   | 0.98 (0.84-1.16)       | 1.11 (0.93-1.33) |
| Death                       | 250                                              | 273                                   | 191                        | 1                                   | 1.13 (0.94-1.36)       | 1.21 (0.98-1.49) |
| HF hospitalization          | 162                                              | 186                                   | 125                        | 1                                   | 0.95 (0.74-1.21)       | 1.06 (0.81-1.38) |

<sup>a</sup>Multivariable analyses were adjusted for age, sex, inpatient recruitment, body mass index, education, current smoker, alcohol intake, NYHA functional class, history of diabetes, history of chronic obstructive pulmonary disease, beta-blocker use, renin-angiotensin system inhibitor use, mineralocorticoid receptor antagonist use, implantable cardioverter-defibrillator use, stage 4 or 5 kidney disease, left ventricular ejection fraction category, HF etiology, systolic blood pressure, self-reported low-salt diet, and geographical region. Analyses were also adjusted for the following dietary variables (measured in grams per day): For associations with unprocessed red meats, models are adjusted for fruits, vegetables, legumes, and starches. For other associations, models were adjusted for fruits, vegetables, legumes, starches, and unprocessed red meats. <sup>b</sup>No associations were statistically significant at a threshold value of  $P < 0.05$ .

Abbreviations as in [Table 2](#).

(HR: 0.77; 95% CI: 0.61-0.97) compared with <1 serving per day. Higher refined grain intake was not significantly associated with the composite primary outcome, but it was associated with a higher risk of hospitalization for HF (HR: 1.56 [95% CI: 1.19-2.05] for 1 to <3 servings per day and HR: 1.76 [95% CI: 1.30-2.39] for ≥3 servings per day vs <1 serving per day). Associations between legumes, vegetables, and

refined grains and HF hospitalization remained consistent in sensitivity analyses that accounted for the competing risk of death ([Supplemental Table 3](#)). Other plant-based foods had neutral associations with the primary outcome and with its components. None of the animal-based foods studied were significantly associated with the primary outcome or its components.

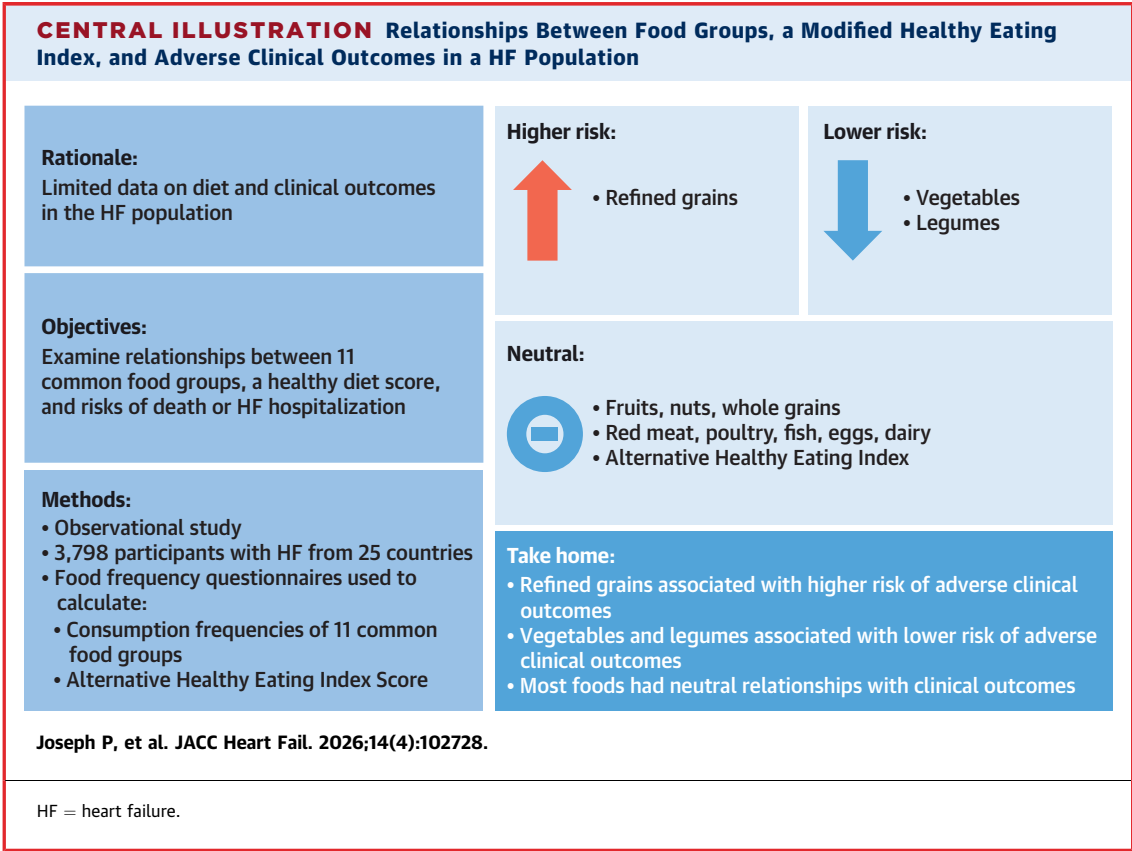
**TABLE 4 Associations Between the Modified AHEI and Clinical Outcomes**

|                             | Quartiles and Crude Numbers of Events |                        |                      |                      | Adjusted HR (95% CI) <sup>a,b</sup> |                  |                  |                  |
|-----------------------------|---------------------------------------|------------------------|----------------------|----------------------|-------------------------------------|------------------|------------------|------------------|
|                             | AHEI Q1<br>(n = 952)                  | AHEI Q2<br>(n = 1,066) | AHEI Q3<br>(n = 833) | AHEI Q4<br>(n = 947) | AHEI Q1                             | AHEI Q2          | AHEI Q3          | AHEI Q4          |
| Death or HF hospitalization | 297                                   | 332                    | 225                  | 293                  | 1                                   | 1.05 (0.88-1.25) | 0.91 (0.75-1.12) | 0.87 (0.71-1.07) |
| Death                       | 232                                   | 249                    | 167                  | 241                  | 1                                   | 0.99 (0.81-1.21) | 0.90 (0.72-1.14) | 0.91 (0.72-1.15) |
| HF hospitalization          | 134                                   | 147                    | 95                   | 97                   | 1                                   | 1.13 (0.87-1.46) | 0.91 (0.67-1.24) | 0.79 (0.57-1.09) |

<sup>a</sup>Analyses adjusted for age, sex, inpatient recruitment, body mass index, education, current smoker, alcohol intake, NYHA functional class, history of diabetes, history of chronic obstructive pulmonary disease, beta-blocker use, renin-angiotensin system inhibitor use, mineralocorticoid receptor antagonist use, implantable cardioverter-defibrillator use, stage 4 or 5 kidney disease, left ventricular ejection fraction category, HF etiology, systolic blood pressure, self-reported low-salt diet, and geographical region. <sup>b</sup>No associations were statistically significant at a threshold value of  $P < 0.05$ .

Abbreviations as in [Tables 1 and 2](#).





Associations between mAHEI scores and clinical outcomes are summarized in [Table 4](#). Compared with the lowest quartile of mAHEI scores, higher scores were not associated with the primary outcome, or its components.

DISCUSSION

This large multinational cohort study from 25 countries found that some components of diet may be important modifiable determinants for adverse clinical outcomes in persons with HF ([Central Illustration](#)). Higher legume intake was shown to be associated with a lower risk of the combined outcome of a death or a HF hospitalization, moderate vegetable intake was associated with a lower risk of HF hospitalization, and higher refined grain intake was associated with a higher risk of HF hospitalization. These findings are generally consistent with the associations between these food groups and cardiovascular events observed in the general population.<sup>11,16</sup> The effects of these foods were largely driven by associations with HF hospitalizations as opposed to deaths. Finally, most foods had neutral effects on clinical outcomes, which should

also be emphasized when providing dietary guidance to persons with HF.

Compliance to dietary advice is generally poor in the HF population, and one reason for this may be the lack of focused and practical guidance.<sup>17,18</sup> Although clinical trials are the ideal studies for the basis of recommendations, sufficiently powered trials testing dietary interventions in HF populations are lacking. One exception was the large SODIUM-HF (Study of Dietary Intervention Under 100 MMOL in Heart Failure) trial, which effectively established no benefit from restricting sodium consumption to <2.5 g/d in patients with HF.<sup>3</sup> Otherwise, trials such as the GOURMET-HF (Geriatric Out-of-Hospital Randomized Meal Trial in Heart Failure) trial (which randomized 66 patients with HF to a DASH [Dietary Approaches to Stop Hypertension] diet or to their usual diet) have been too small to use as the basis for definitive recommendations.<sup>19</sup> In this context, the findings of the current study can be used to inform practical guidance on how specific food choices affect the clinical course of HF.

In general populations, measures of a healthy dietary pattern such as the AHEI are associated with a lower risk of cardiovascular disease, but prior data

specific to HF populations were limited.<sup>20–22</sup> One study of 991 patients with HF enrolled during an acute HF hospitalization reported a 26% proportional lower risk of rehospitalization in those adherent to a Mediterranean diet.<sup>23</sup> The Women's Health Initiative study reported that in participants who developed HF during follow-up, greater adherence to the DASH diet at baseline was associated with a 16% lower proportional risk of subsequent death.<sup>24</sup> The current sample population is more generalizable to the global HF population compared with these earlier studies, and we observed no significant association between overall diet quality and death or HF hospitalization. These findings are also largely consistent with our results for individual food groups, in which most foods studied had neutral associations with these clinical outcomes.

**STUDY LIMITATIONS.** FFQs are the most used method to collect dietary data in nutrition epidemiologic studies but are better suited for ranking individuals as opposed to a measure of absolute intake. Some measurement error related to dietary recall is also inevitable. Because the FFQs were mainly focused on the common food groups consumed within each country, we were not able to estimate total energy or nutrient intake and we were unable to account for this as a covariate in the regression models. Up-to-date nutrient databases with information on trans fatty acids, particularly from industrially produced sources, is limited for most countries, and the AHEI has been commonly modified to exclude this component. Regional groupings would not account for subregional differences in dietary patterns that may affect HF outcomes. Our analysis focused on food groups that are commonly consumed across regions. Understanding the effects of dietary patterns unique to a region will require larger studies of HF populations within that region.

In the current study, the cause of a death or a hospitalization was determined by the local investigator based on the available clinical data. This could have led to some misclassification of first HF hospitalization but would not affect all-cause mortality. Our analyses did not account for multiple hypothesis testing for the adjustment of *P* values or CIs. Although the current study is among the largest to examine relationships between dietary factors and clinical outcomes in the HF population, it may still be underpowered to detect small effect sizes. Finally,

although extensive adjustments for potential confounding factors were performed in the regression models, residual confounding is still possible. These limitations need to be considered when interpreting our findings. Future studies in HF populations are important to both validate our findings and to identify the underlying mechanisms of these observed effects.

## CONCLUSIONS

This study found that some foods may be important modifiable determinants of adverse outcomes in the HF population. The findings support higher consumption of legumes and vegetables, and lower consumption of refined grains, in persons with HF.

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## PERSPECTIVES

### COMPETENCY IN PATIENT CARE AND

**PROCEDURAL SKILLS:** In persons with HF, vegetable and legume consumption were each associated with a lower risk of adverse clinical outcomes, whereas refined grain consumption was associated with higher risk. Most dietary factors had neutral relationships with clinical

outcomes. These findings add to the current knowledge of optimizing diet in persons with HF.

**TRANSLATIONAL OUTLOOK:** These findings can inform dietary guidelines in persons with HF. Performance of future validation studies is also important.

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**KEY WORDS** clinical outcomes, diet, heart failure, registry

**APPENDIX** For supplemental tables, please see the online version of this paper.



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